

四、分部积分法

问题 $\int x \cos x dx = ?$

解决思路 利用两个函数乘积的求导法则.

设函数 $f(x)$ 和 $g(x)$ 具有连续导数,

$$d(f(x)g(x)) = f(x)dg(x) + g(x)df(x)$$

$$\int f(x)dg(x) = f(x)g(x) - \int g(x)df(x).$$

分部积分公式



例33 求积分 $\int x \cos x dx$.

解 (一) 令 $f(x) = \cos x$, $x dx = \frac{1}{2} dx^2 = dg(x)$

$$\int x \cos x dx = \int \cos x d \frac{x^2}{2} = \frac{x^2}{2} \cos x + \int \frac{x^2}{2} \sin x dx$$

显然, $f(x), g'(x)$ 选择不当, 积分更难进行.

解 (二) 令 $f(x) = x$, $\cos x dx = d \sin x = df(x)$

$$\begin{aligned} \int x \cos x dx &= \int x d \sin x = x \sin x - \int \sin x dx \\ &= x \sin x + \cos x + C. \end{aligned}$$

反对幂三指



例35 求积分 $\int x \arctan x dx$.

解 令 $f(x) = \arctan x$, $xdx = d\frac{x^2}{2} = dg(x)$

$$\begin{aligned}\int x \arctan x dx &= \int \arctan x d\frac{x^2}{2} \\&= \frac{x^2}{2} \arctan x - \int \frac{x^2}{2} d(\arctan x) \\&= \frac{x^2}{2} \arctan x - \int \frac{x^2}{2} \cdot \frac{1}{1+x^2} dx \\&= \frac{x^2}{2} \arctan x - \int \frac{1}{2} \cdot \left(1 - \frac{1}{1+x^2}\right) dx \\&= \frac{x^2}{2} \arctan x - \frac{1}{2} (x - \arctan x) + C.\end{aligned}$$



例36 求积分 $\int \ln x dx$.

例38 求积分 $\int e^x \sin x dx$.

解 $\int e^x \sin x dx = \int \sin x de^x$

$$= e^x \sin x - \int e^x d(\sin x)$$
$$= e^x \sin x - \int e^x \cos x dx = e^x \sin x - \int \cos x de^x$$
$$= e^x \sin x - (e^x \cos x - \int e^x d \cos x)$$
$$= e^x (\sin x - \cos x) - \int e^x \sin x dx$$

注意循环形式

$$\therefore \int e^x \sin x dx = \frac{e^x}{2} (\sin x - \cos x) + C.$$



例39 求积分 $\int \frac{x \arctan x}{\sqrt{1+x^2}} dx$.

解
$$\begin{aligned} \int \frac{x \arctan x}{\sqrt{1+x^2}} dx &= \frac{1}{2} \int \frac{\arctan x}{\sqrt{1+x^2}} dx^2 \\ &= \frac{1}{2} \int \frac{\arctan x}{\sqrt{1+x^2}} d(1+x^2) = \int \arctan x d\sqrt{1+x^2} \\ &= \sqrt{1+x^2} \arctan x - \int \sqrt{1+x^2} d(\arctan x) \\ &= \sqrt{1+x^2} \arctan x - \int \sqrt{1+x^2} \cdot \frac{1}{1+x^2} dx \end{aligned}$$



$$= \sqrt{1+x^2} \arctan x - \int \frac{1}{\sqrt{1+x^2}} dx \quad \text{令 } x = \tan t$$

$$\int \frac{1}{\sqrt{1+x^2}} dx = \int \frac{1}{\sqrt{1+\tan^2 t}} \sec^2 t dt = \int \sec t dt$$

$$= \ln(\sec t + \tan t) + C = \ln(x + \sqrt{1+x^2}) + C$$

$$\therefore \int \frac{x \arctan x}{\sqrt{1+x^2}} dx$$

$$= \sqrt{1+x^2} \arctan x - \ln(x + \sqrt{1+x^2}) + C.$$



例42 已知 $f(x)$ 的一个原函数是 e^{-x^2} , 求 $\int xf'(x)dx$.

$$\text{解 } \int xf'(x)dx = \int xdf(x) = xf(x) - \int f(x)dx,$$

$$\because \left(\int f(x)dx \right)' = f(x), \quad \therefore \int f(x)dx = e^{-x^2} + C,$$

$$\text{两边同时对 } x \text{ 求导, 得 } f(x) = -2xe^{-x^2},$$

$$\begin{aligned} \therefore \int xf'(x)dx &= xf(x) - \int f(x)dx \\ &= -2x^2e^{-x^2} - e^{-x^2} + C. \end{aligned}$$



五、有理函数的积分

有理函数的定义：

两个多项式的商表示的函数称之.

$$\frac{P(x)}{Q(x)} = \frac{a_0 x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n}{b_0 x^m + b_1 x^{m-1} + \cdots + b_{m-1} x + b_m}$$

其中 m 、 n 都是非负整数； $a_0, a_1, \cdots, a_n$ 及
 $b_0, b_1, \cdots, b_m$ 都是实数，并且 $a_0 \neq 0$ ， $b_0 \neq 0$.



假定分子与分母之间没有公因式

(1) $n < m$, 这有理函数是**真分式**;

(2) $n \geq m$, 这有理函数是**假分式**;

利用多项式除法, 假分式可以化成一个多项式和一个真分式之和.

例
$$\frac{x^3 + x + 1}{x^2 + 1} = x + \frac{1}{x^2 + 1}.$$

难点 将有理真分式函数化为部分分式之和.

$$Q_m(x) = b_0(x-a)^\alpha \cdots (x-b)^\beta (x^2 + px + q)^\lambda \cdots (x^2 + rx + s)^\mu$$



有理函数化为部分分式之和的一般规律：

(1) 分母中若有因式 $(x-a)^k$ ，则分解后为

$$\frac{A_1}{(x-a)^k} + \frac{A_2}{(x-a)^{k-1}} + \cdots + \frac{A_k}{x-a},$$

其中 $A_1, A_2, \cdots, A_k$ 都是常数.

特殊地： $k=1$, 分解后为 $\frac{A}{x-a}$;



(2) 分母中若有因式 $(x^2 + px + q)^k$, 其中 $p^2 - 4q < 0$ 则分解后为

$$\frac{M_1x + N_1}{(x^2 + px + q)^k} + \frac{M_2x + N_2}{(x^2 + px + q)^{k-1}} + \cdots + \frac{M_kx + N_k}{x^2 + px + q}$$

其中 M_i, N_i 都是常数 ($i = 1, 2, \dots, k$).

特殊地: $k = 1$, 分解后为 $\frac{Mx + N}{x^2 + px + q}$;



真分式化为部分分式之和的待定系数法

例41
$$\frac{x+3}{x^2-5x+6} = \frac{x+3}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3},$$

$$\therefore x+3 = A(x-3) + B(x-2),$$

$$\therefore x+3 = (A+B)x - (3A+2B),$$

$$\Rightarrow \begin{cases} A+B=1, \\ -(3A+2B)=3, \end{cases} \Rightarrow \begin{cases} A=-5 \\ B=6 \end{cases},$$

$$\therefore \frac{x+3}{x^2-5x+6} = \frac{-5}{x-2} + \frac{6}{x-3}.$$



例42
$$\frac{1}{x(x-1)^2} = \frac{A}{x} + \frac{B}{(x-1)^2} + \frac{C}{x-1},$$

$$1 = A(x-1)^2 + Bx + Cx(x-1) \quad (1)$$

代入特殊值来确定系数 A, B, C

取 $x = 0, \Rightarrow A = 1$ 取 $x = 1, \Rightarrow B = 1$

取 $x = 2$, 并将 A, B 值代入(1) $\Rightarrow C = -1$

$$\therefore \frac{1}{x(x-1)^2} = \frac{1}{x} + \frac{1}{(x-1)^2} - \frac{1}{x-1}.$$



例43
$$\frac{1}{(1+2x)(1+x^2)} = \frac{A}{1+2x} + \frac{Bx+C}{1+x^2},$$

$$1 = A(1+x^2) + (Bx+C)(1+2x),$$

整理得 $1 = (A+2B)x^2 + (B+2C)x + C + A,$

$$\begin{cases} A+2B=0, \\ B+2C=0, \\ A+C=1, \end{cases} \Rightarrow A = \frac{4}{5}, B = -\frac{2}{5}, C = \frac{1}{5},$$

$$\therefore \frac{1}{(1+2x)(1+x^2)} = \frac{\frac{4}{5}}{1+2x} + \frac{-\frac{2}{5}x + \frac{1}{5}}{1+x^2}.$$



例44 求积分 $\int \frac{1}{x(x-1)^2} dx$.

解
$$\int \frac{1}{x(x-1)^2} dx = \int \left[\frac{1}{x} + \frac{1}{(x-1)^2} - \frac{1}{x-1} \right] dx$$

$$= \int \frac{1}{x} dx + \int \frac{1}{(x-1)^2} dx - \int \frac{1}{x-1} dx$$

$$= \ln|x| - \frac{1}{x-1} - \ln|x-1| + C.$$



例45 求积分 $\int \frac{1}{(1+2x)(1+x^2)} dx$.

$$\begin{aligned}\text{解 } \int \frac{1}{(1+2x)(1+x^2)} dx &= \int \frac{\frac{4}{5}}{1+2x} dx + \int \frac{-\frac{2}{5}x + \frac{1}{5}}{1+x^2} dx \\ &= \frac{2}{5} \int \frac{1}{1+2x} d(1+2x) - \frac{1}{5} \int \frac{2x}{1+x^2} dx + \frac{1}{5} \int \frac{1}{1+x^2} dx \\ &= \frac{2}{5} \ln|1+2x| - \frac{1}{5} \ln(1+x^2) + \frac{1}{5} \arctan x + C.\end{aligned}$$



说明 将有理函数化为部分分式之和后，只出现三类情况：

(1) 多项式； (2) $\frac{A}{(x-a)^n}$ ； (3) $\frac{Mx+N}{(x^2+px+q)^n}$ ；

讨论积分 $\int \frac{Mx+N}{(x^2+px+q)^n} dx$,

$$\because x^2 + px + q = \left(x + \frac{p}{2} \right)^2 + q - \frac{p^2}{4},$$
$$\text{令 } x + \frac{p}{2} = t$$



$$\text{记 } x^2 + px + q = t^2 + a^2, \quad Mx + N = Mt + b,$$

$$\text{则 } a^2 = q - \frac{p^2}{4}, \quad b = N - \frac{Mp}{2},$$

$$\therefore \int \frac{Mx + N}{(x^2 + px + q)^n} dx$$

$$= \int \frac{Mt}{(t^2 + a^2)^n} dt + \int \frac{b}{(t^2 + a^2)^n} dt$$



$$(1) \quad n=1, \quad \int \frac{Mx + N}{x^2 + px + q} dx$$

$$= \frac{M}{2} \ln(x^2 + px + q) + \frac{b}{a} \arctan \frac{x + \frac{p}{2}}{a} + C;$$

$$(2) \quad n > 1, \quad \int \frac{Mx + N}{(x^2 + px + q)^n} dx$$

$$= -\frac{M}{2(n-1)(t^2 + a^2)^{n-1}} + b \int \frac{1}{(t^2 + a^2)^n} dt.$$

P292, 例49: 利用分部积分法, 计算递推公式。

这三类积分均可积出, 且原函数都是初等函数.

结论 有理函数的原函数都是初等函数.



六、三角函数有理式的积分

三角有理式的定义：

由三角函数和常数经过有限次四则运算构成的函数称之。一般记为 $R(\sin x, \cos x)$

$$\because \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2} = \frac{2 \tan \frac{x}{2}}{\sec^2 \frac{x}{2}} = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}},$$

$$\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2},$$



$$\cos x = \frac{1 - \tan^2 \frac{x}{2}}{\sec^2 \frac{x}{2}} = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}},$$

令 $u = \tan \frac{x}{2}$ $x = 2\arctan u$ (万能置换公式)

$$\sin x = \frac{2u}{1+u^2}, \quad \cos x = \frac{1-u^2}{1+u^2}, \quad dx = \frac{2}{1+u^2} du$$

$$\int R(\sin x, \cos x) dx = \int R\left(\frac{2u}{1+u^2}, \frac{1-u^2}{1+u^2}\right) \frac{2}{1+u^2} du.$$



例46 求积分 $\int \frac{\sin x}{1 + \sin x + \cos x} dx$.

解 由万能置换公式 $\sin x = \frac{2u}{1+u^2}$,

$$\cos x = \frac{1-u^2}{1+u^2} \quad dx = \frac{2}{1+u^2} du,$$

$$\begin{aligned} \int \frac{\sin x}{1 + \sin x + \cos x} dx &= \int \frac{2u}{(1+u)(1+u^2)} du \\ &= \int \frac{2u + 1 + u^2 - 1 - u^2}{(1+u)(1+u^2)} du \end{aligned}$$



$$= \int \frac{(1+u)^2 - (1+u^2)}{(1+u)(1+u^2)} du = \int \frac{1+u}{1+u^2} du - \int \frac{1}{1+u} du$$

$$= \arctan u + \frac{1}{2} \ln(1+u^2) - \ln|1+u| + C$$

$$\because u = \tan \frac{x}{2}$$

$$= \frac{x}{2} + \ln \left| \sec \frac{x}{2} \right| - \ln \left| 1 + \tan \frac{x}{2} \right| + C.$$



例47 求积分 $\int \frac{1}{\sin^4 x} dx$.

解 (一) $u = \tan \frac{x}{2}$, $\sin x = \frac{2u}{1+u^2}$, $dx = \frac{2}{1+u^2} du$,

$$\int \frac{1}{\sin^4 x} dx = \int \frac{1+3u^2+3u^4+u^6}{8u^4} du$$

$$= \frac{1}{8} \left[-\frac{1}{3u^3} - \frac{3}{u} + 3u + \frac{u^3}{3} \right] + C$$

$$= -\frac{1}{24 \left(\tan \frac{x}{2} \right)^3} - \frac{3}{8 \tan \frac{x}{2}} + \frac{3}{8} \tan \frac{x}{2} + \frac{1}{24} \left(\tan \frac{x}{2} \right)^3 + C.$$



解（二） 修改万能置换公式，令 $u = \tan x$

$$\sin x = \frac{u}{\sqrt{1+u^2}}, \quad dx = \frac{1}{1+u^2} du,$$

$$\int \frac{1}{\sin^4 x} dx = \int \frac{1}{\left(\frac{u}{\sqrt{1+u^2}}\right)^4} \cdot \frac{1}{1+u^2} du = \int \frac{1+u^2}{u^4} du$$

$$= -\frac{1}{3u^3} - \frac{1}{u} + C = -\frac{1}{3} \cot^3 x - \cot x + C.$$



解（三） 可以不用万能置换公式.

$$\begin{aligned}\int \frac{1}{\sin^4 x} dx &= \int \csc^2 x (1 + \cot^2 x) dx \\ &= \int \csc^2 x dx + \int \cot^2 x \boxed{\csc^2 x dx} = d(\cot x) \\ &= -\cot x - \frac{1}{3} \cot^3 x + C.\end{aligned}$$

结论 比较以上三种解法, 便知万能置换不一定是最佳方法, 故三角有理式的计算中先考虑其它手段, 不得已才用万能置换.



例48 求积分 $\int \frac{1 + \sin x}{\sin 3x + \sin x} dx$.

解 $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$

$$\begin{aligned} \int \frac{1 + \sin x}{\sin 3x + \sin x} dx &= \int \frac{1 + \sin x}{2 \sin 2x \cos x} dx \\ &= \int \frac{1 + \sin x}{4 \sin x \cos^2 x} dx \\ &= \frac{1}{4} \int \frac{1}{\sin x \cos^2 x} dx + \frac{1}{4} \int \frac{1}{\cos^2 x} dx \end{aligned}$$



$$= \frac{1}{4} \int \frac{\sin^2 x + \cos^2 x}{\sin x \cos^2 x} dx + \frac{1}{4} \int \frac{1}{\cos^2 x} dx$$

$$= \frac{1}{4} \int \frac{\sin x}{\cos^2 x} dx + \frac{1}{4} \int \frac{1}{\sin x} dx + \frac{1}{4} \int \frac{1}{\cos^2 x} dx$$

$$= -\frac{1}{4} \int \frac{1}{\cos^2 x} d(\cos x) + \frac{1}{4} \int \frac{1}{\sin x} dx + \frac{1}{4} \int \frac{1}{\cos^2 x} dx$$

$$= \frac{1}{4 \cos x} + \frac{1}{4} \ln \left| \tan \frac{x}{2} \right| + \frac{1}{4} \tan x + C.$$



七、简单无理函数的积分

讨论类型 $R(x, \sqrt[n]{ax+b}), R(x, \sqrt[n]{\frac{ax+b}{cx+e}}),$

解决方法 作代换去掉根号.

例49 求积分 $\int \frac{1}{x} \sqrt{\frac{1+x}{x}} dx$

解 令 $\sqrt{\frac{1+x}{x}} = t \Rightarrow \frac{1+x}{x} = t^2,$



$$x = \frac{1}{t^2 - 1}, \quad dx = -\frac{2t dt}{(t^2 - 1)^2},$$

$$\int \frac{1}{x} \sqrt{\frac{1+x}{x}} dx = -\int (t^2 - 1)t \frac{2t}{(t^2 - 1)^2} dt = -2 \int \frac{t^2 dt}{t^2 - 1}$$

$$= -2 \int \left(1 + \frac{1}{t^2 - 1} \right) dt = -2t - \ln \frac{t-1}{t+1} + C$$

$$= -2 \sqrt{\frac{1+x}{x}} - \ln \left[x \left(\sqrt{\frac{1+x}{x}} - 1 \right)^2 \right] + C.$$



例50 求 $\int \frac{1}{\sqrt{1+e^x}} dx$.

解 令 $t = \sqrt{1+e^x} \Rightarrow e^x = t^2 - 1$,

$$x = \ln(t^2 - 1), \quad dx = \frac{2t}{t^2 - 1} dt,$$

$$\int \frac{1}{\sqrt{1+e^x}} dx = \int \frac{2}{t^2 - 1} dt = \int \left(\frac{1}{t-1} - \frac{1}{t+1} \right) dt$$

$$= \ln \left| \frac{t-1}{t+1} \right| + C = 2 \ln(\sqrt{1+e^x} - 1) - x + C.$$



例51 求 $\int \frac{1}{\sqrt{x}(1+\sqrt[3]{x})} dx$.

解 令 $x = t^6 \Rightarrow dx = 6t^5 dt$,

$$\int \frac{1}{\sqrt{x}(1+\sqrt[3]{x})} dx = \int \frac{6t^5}{t^3(1+t^2)} dt$$

$$= \int \frac{6t^2}{1+t^2} dt = 6 \int \frac{t^2 + 1 - 1}{1+t^2} dt$$

$$= 6 \int \left(1 - \frac{1}{1+t^2} \right) dt = 6[t - \arctan t] + C$$

$$= 6[\sqrt[6]{x} - \arctan \sqrt[6]{x}] + C.$$



例52 求积分 $\int \frac{1}{\sqrt{x+1} + \sqrt[3]{x+1}} dx$.

解 令 $t^6 = x+1 \Rightarrow 6t^5 dt = dx$,

$$\begin{aligned} \int \frac{1}{\sqrt{x+1} + \sqrt[3]{x+1}} dx &= \int \frac{1}{t^3 + t^2} \cdot 6t^5 dt \\ &= 6 \int \frac{t^3}{t+1} dt = 2t^3 - 3t^2 + 6t + 6\ln|t+1| + C \\ &= 2\sqrt{x+1} - 3\sqrt[3]{x+1} + 3\sqrt[6]{x+1} + 6\ln(\sqrt[6]{x+1} + 1) + C. \end{aligned}$$

说明 无理函数去根号时, 取根指数的**最小公倍数**.



例53 求积分 $\int \frac{x}{\sqrt{3x+1} + \sqrt{2x+1}} dx.$

解 先对分母进行有理化

$$\begin{aligned}\text{原式} &= \int \frac{x(\sqrt{3x+1} - \sqrt{2x+1})}{(\sqrt{3x+1} + \sqrt{2x+1})(\sqrt{3x+1} - \sqrt{2x+1})} dx \\&= \int (\sqrt{3x+1} - \sqrt{2x+1}) dx \\&= \frac{1}{3} \int \sqrt{3x+1} d(3x+1) - \frac{1}{2} \int \sqrt{2x+1} d(2x+1) \\&= \frac{2}{9} (3x+1)^{\frac{3}{2}} - \frac{1}{3} (2x+1)^{\frac{3}{2}} + C.\end{aligned}$$



例54 求积分 $\int \frac{1}{1 + e^{\frac{x}{2}} + e^{\frac{x}{3}} + e^{\frac{x}{6}}} dx.$

解 令 $t = e^{\frac{x}{6}} \Rightarrow x = 6 \ln t, \quad dx = \frac{6}{t} dt,$

$$\int \frac{1}{1 + e^{\frac{x}{2}} + e^{\frac{x}{3}} + e^{\frac{x}{6}}} dx = \int \frac{1}{1 + t^3 + t^2 + t} \cdot \frac{6}{t} dt$$

$$= 6 \int \frac{1}{t(1+t)(1+t^2)} dt = \int \left(\frac{6}{t} - \frac{3}{1+t} - \frac{3t+3}{1+t^2} \right) dt$$



$$\begin{aligned}
&= \int \left(\frac{6}{t} - \frac{3}{1+t} - \frac{3t+3}{1+t^2} \right) dt \\
&= 6 \ln t - 3 \ln(1+t) - \frac{3}{2} \int \frac{d(1+t^2)}{1+t^2} - 3 \int \frac{1}{1+t^2} dt \\
&= 6 \ln t - 3 \ln(1+t) - \frac{3}{2} \ln(1+t^2) - 3 \arctan t + C \\
&= x - 3 \ln(1 + e^{\frac{x}{6}}) - \frac{3}{2} \ln(1 + e^{\frac{x}{3}}) - 3 \arctan(e^{\frac{x}{6}}) + C.
\end{aligned}$$

